

5 Number Summary.

1. Minimum

2. 1st Quartile

3. Median

4. 3rd Quartile

5. Maximum

Eg: 1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6,
7, 8, 8, 9, 27 outliers

[Lower fence $\longleftrightarrow$ Higher fence]

$$\text{Lower fence} = Q_1 - 1.5(IQR)$$

$$\text{Higher fence} = Q_3 - 1.5(IQR)$$

25% $Q_1 = \frac{\text{Percentile} \times (n+1)}{100} = \frac{25 \times 20}{100} = 5^{\text{th}} \text{ position} = 3$

75% $Q_3 = \frac{75 \times 20}{100} = 15^{\text{th}} \text{ position} = 7$

$$IQR = Q_3 - Q_1 = 7 - 3 = 4$$

$$\text{Lower fence} = Q_1 - 1.5(IQR)$$

$$= 3 - 1.5(4)$$

$$= 3 - 6 = -3$$

outliers
outliers
[-3 ↔ 13]

$$\text{Higher fence} = Q_3 + 1.5(IQR)$$

$$= 7 + 1.5(4)$$

$$= 7 + 6 = 13$$

⇒ 1, 2, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, 27

Minimum = 1

1st Quartile = 3

Median = 5

3rd Quartile = 7

Maximum = 9

Box Plot.